

TEAM HYPERION

Brisbane Boys' College - Open Soccer League - Queensland State Competition



Our Team (Left to Right)

Thomas McCabe - Sensors/Software
Luke Atherton - Structural/Hardware
Sam Garg - Strategy/Software
Matthew Adams - Electrical/Hardware

Our Innovation - Light Sensor Board

The Light Sensor Board features 48 distinct component groups, each equipped with phototransistors, LEDs, resistors, and smoothing capacitors. These sensors are strategically arranged in a circular pattern ahead of the wheels to optimize a defender's positioning near the goal box, while 16 outer groups are divided into four segments between the wheels to ensure rapid line detection and response. To manage this dense array efficiently, all sensor signals are routed through three 16-bit multiplexers. This hardware configuration successfully consolidates the multiplexed data into just four analog pins, significantly reducing the number of processor pins required for operation.

To find the direction of the line, the robot is simply trying to find the most accurate "middle" of whatever it currently sees. Depending on how many pieces (clusters) of the line the robot detects, here is what it does:

- One Cluster: The line is straight at that cluster! The direction is just the center of that single piece.
- Two Clusters: The line is in the middle of that cluster! The robot pinpoints exactly halfway between the two pieces.
- Three Clusters: It completes a balancing act, where it finds the middle pieces and the two outside pieces. It calculates the halfway point between the middle and left side, and then the middle and right side. After doing this, it averages those two paths to get a smooth and balanced direction.

Our Challenge - Mechanical Stability, Traction and Weight & Kicker Designing

The Challenge: The robot struggled with traction on the slippery felt field, and there was a risk of the wheels sliding off the motor axles. Additionally, upgrading the robot with a new kicker, dribbler, and dual-layer wheels added unwanted mass (as we were previously in the Infrared Division, which had a strict weight limit).

The Solution: The team engineered custom dual-layered omni wheels with a specialised wheel hub. This hub features an internal groove for a square nut, allowing a bolt to tighten directly onto the motor's D-shaft for a highly secure connection. To offset the added weight of the new components, they optimized their existing parts and internal layout to minimize mass while maintaining structural strength.

The Challenge: Delivering a consistently powerful kick while ensuring reliable contact with the ball.

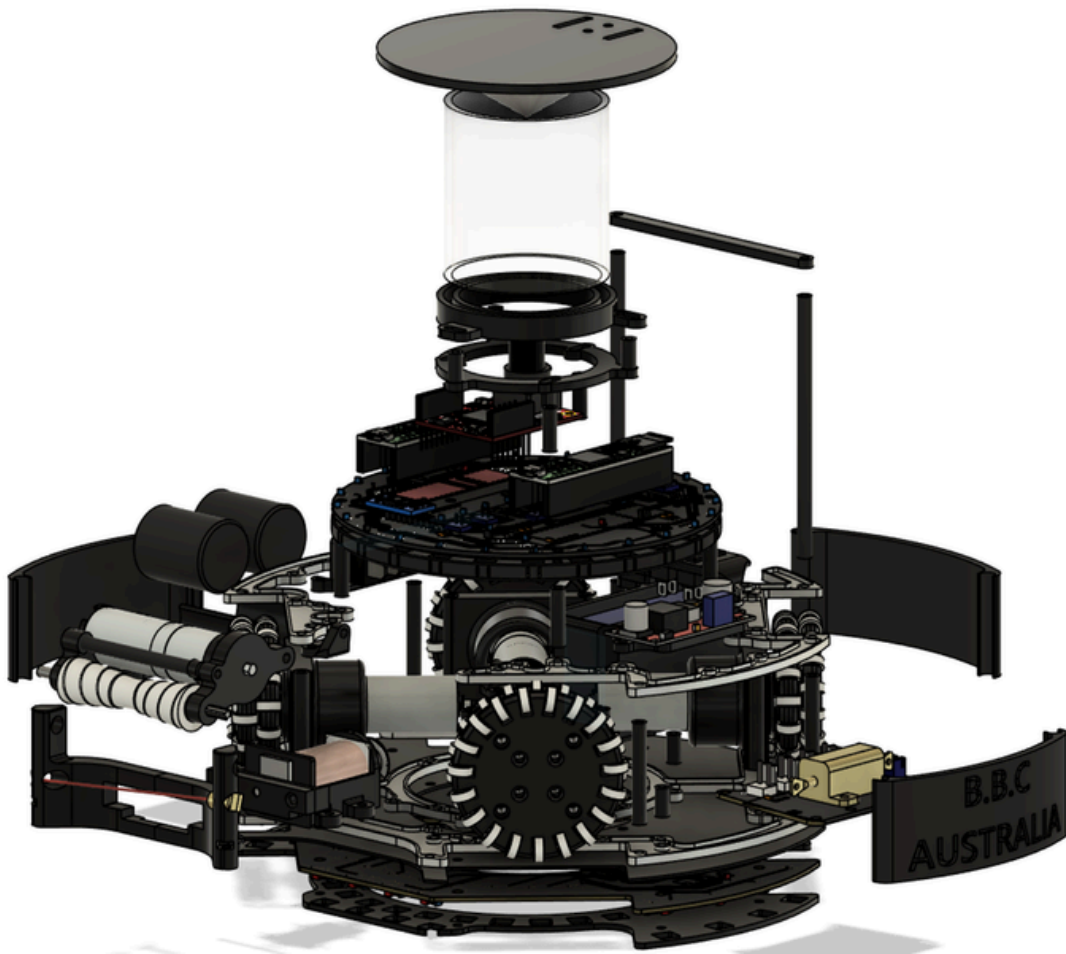
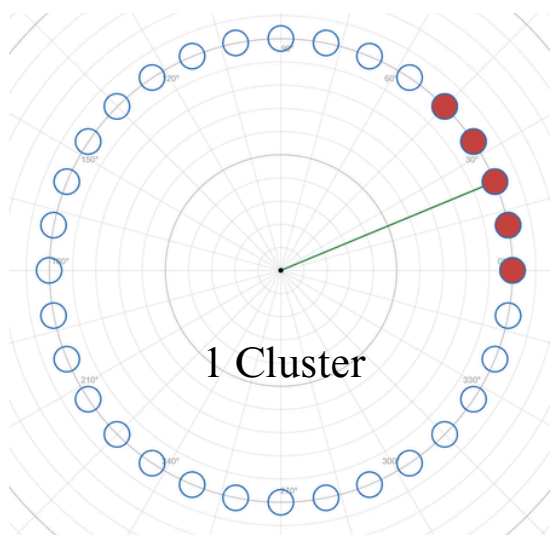
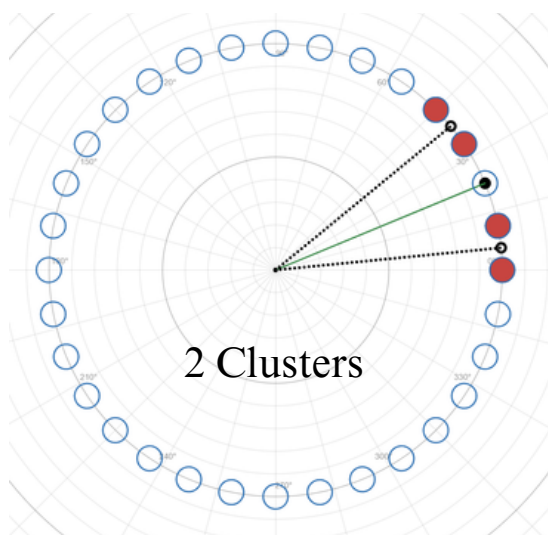
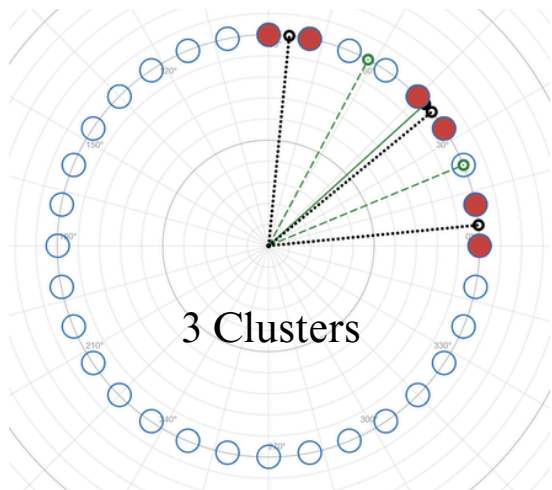
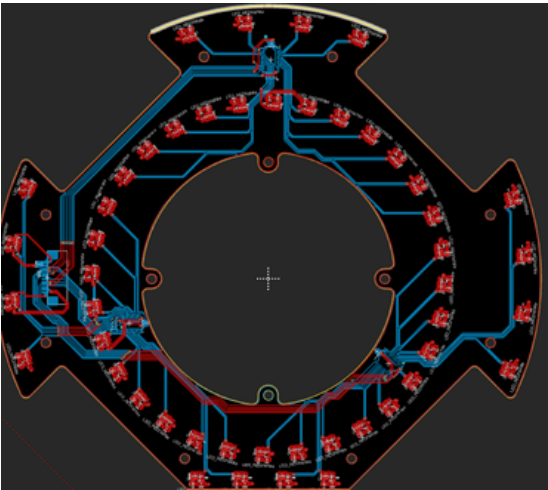
The Solution: The team heavily modified a TAKAHA CB1037 solenoid. They wound a custom 250-turn coil and replaced the standard tip with a horizontal bar to ensure reliable ball contact. To achieve the necessary power, they designed a circuit that boosts their 12V battery supply to 48V, charging two parallel 4700µF capacitors to discharge a massive burst of energy.

Who are we?

We are Hyperion, a four-student robotics team from Brisbane Boys' College, and right now, our primary focus is gearing up for the Queensland Competition in the Open Soccer League. We have just returned from representing Australia at the 2026 RoboCup in Incheon, South Korea, where we secured 12th Place Individually and 4th Place in SuperTeam, which proudly builds on our 2025 run in Salvador, Brazil (14th Individually, 1st Place SuperTeam). Coming off those international campaigns in the Infrared league, we are currently executing rapid technical modifications to our robot system tweaking our 360° vision processing and movement algorithms to handle the new field conditions of the Open league. Despite the quick turnaround, we are incredibly excited to hit the state field and remain dedicated to sharing our open-source designs and workflows to help future teams master these advanced autonomous systems.

Previous Competition Experience

- 2026 Internationals - South Korea
- 2026 Brisbane Regionals
- 2025 Australian Nationals - Canberra
- 2025 Queensland States
- 2025 Internationals - Brazil
- 2024 Australian Nationals - Queensland
- 2024 Queensland States
- 2024 Brisbane Regionals
- + Numerous Individual RoboCup Competition Experience



What have we learnt throughout our time at RoboCup?

Throughout our time at RoboCup, we have experienced highs and lows. Whether that was the unparalleled thrill of placing in a competition, or the absolute panic of our robot's wheels disintegrating right before a match, we have experienced a lot of what RoboCup has to offer. As a result of this, there are many things that we have learnt. This includes valuable technical and coding skills that we can take into our future professions and use to stand out in job applications. Moving beyond basic programming, we evolved into multidisciplinary engineers. Structurally, we learned how to design resilient robots and account for the brutal physical forces our robots would face on the field. Electrically, we tackled power distribution, soldered reliable connections, and ensured clean signal integrity between components. By combining this hardware knowledge with complex software logic to process sensor data in milliseconds, we truly learned how to bridge the gap between digital code and the unpredictable physical world. However, the most profound lessons haven't just been technical. We have also grown significantly as a team over the years, which has given us the valuable ability to communicate, collaborate, and thrive under immense pressure. We learned how to divide tasks based on our individual strengths, trust one another's expertise, and navigate the high-stress environment of competition days by turning disagreements over design choices into constructive compromises. Most importantly, watching our hard work occasionally fall apart on the mat taught us true resilience. We learned to treat failures not as personal defeats, but as crucial data points to improve our iterative design. Ultimately, RoboCup was never just about making a robot function; it was a pressure cooker that forged us into better problem solvers and a more cohesive team, building a foundation of grit and synergy that will outlast any tournament.



Our Website